

(*This notebook computes the "naive" Rankin-Selberg with shifts in the "Preliminary" section of the paper*)

(*Hec[m]= $\lambda_{\{\Pi\}}(p^m)$, dHec[n]= $\lambda_{\{\widetilde{\Pi}\}}(p^n)$,
Fou[m,n]= $\lambda_{\{\Pi\}}(p^m, p^n)$ *)

```
In[1]:= SetDirectory[NotebookDirectory[]];
```

```
<< gln.m
```

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⋯ SetDelayed: Tag UpperTriangularMatrixQ in UpperTriangularMatrixQ[m_] is Protected.

GL(n)PACK loaded.

```
In[3]:= Schu[n_, d_, nota_ : a] :=  
SchurPolynomial[Table[nota[i], {i, 1, d}], Join[ConstantArray[0, d - 2], {n}]]
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```
In[4]:= InvRSL[X_, d_, nota1_ : a, nota2_ : b] :=  
Product[Product[(1 - nota1[i] * nota2[j] X), {i, 1, d}], {j, 1, d}]
```

```
In[5]:= R[d_, k_, l_] :=  
Cancel[InvRSL[x, d, a, b] * Sum[Schu[m + k, d, a] * Schu[m + l, d, b] x^m, {m, 0, Infinity}]]
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```
In[6]:= Syma = Join[{a[1]}, Table[Denominator[Schu[m, d, a]], {d, 2, 3}]];
```

```
In[7]:= assu = Element[k, Integers] && k ≥ 2;
```

```
In[8]:= Ru1p1 = {a[1] → Hec[1], a[1]^k_ - a[2]^k_ → Hec[k - 1],  
a[1]^(k + n_) * a[2]^r_ - a[1]^(k + n_) * a[3]^r_ + a[2]^(k + n_) * a[3]^r_ - a[2]^(k + n_) * a[1]^r_ +  
a[3]^(k + n_) * a[1]^r_ - a[3]^(k + n_) * a[2]^r_ → Fou[r - 1, k + n - r - 1]};
```

```
In[9]:= Ru1min = {a[1] → Hec[1], -a[1]^k_ + a[2]^k_ → -Hec[k - 1],  
-a[1]^(k + n_) * a[2]^r_ + a[1]^(k + n_) * a[3]^r_ - a[2]^(k + n_) * a[3]^r_ + a[2]^(k + n_) * a[1]^r_ -  
a[3]^(k + n_) * a[1]^r_ + a[3]^(k + n_) * a[2]^r_ → -Fou[r - 1, k + n - r - 1]};
```

```
In[10]:= OmegaRule2[ω_, ass_ : True] :=  
HoldPattern[c_. a[1]^p_ * a[2]^q_] → With[{s = FullSimplify[Min[p, q], ass]},  
c ω^s a[1]^FullSimplify[p - s, ass] * a[2]^FullSimplify[q - s, ass]];
```

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In[11]:= OmegaRule3[ω_, ass_ : True] := HoldPattern[c_. a[1]^p_ * a[2]^q_ * a[3]^r_] →  
With[{s = FullSimplify[Min[p, q, r], ass]}, c ω^s a[1]^FullSimplify[p - s, ass] *  
a[2]^FullSimplify[q - s, ass] * a[3]^FullSimplify[r - s, ass]];
```

```

In[12]:= coeffsAndMonoms[p_, vars_] :=
  Module[{rules, exps, coeffs, monoms}, rules = CoefficientRules[p, vars];
  exps = Keys[rules];
  coeffs = Values[rules];
  monoms = Times @@ MapThread[Power, {vars, #}] & /@ exps;
  {coeffs, monoms}]

In[13]:= ListCoef[d_, k_, l_] := coeffsAndMonoms[Cancel[R[d, k, l] * Syma[d]], {x}]

In[14]:= AfterSym[d_, k_, l_] :=
  (SymmetricReduction[#, Table[b[i], {i, 1, d}], Join[Table[B[i], {i, 1, d - 1}], {ω}]]][1] & /@
  ListCoef[d, k, l][1] /. ω^r_ -> ω

In[15]:= PartA[d_, k_, l_] := coeffsAndMonoms[#, Join[Table[B[i], {i, 1, d - 1}], {ω}]] & /@ AfterSym[d, k, l]

In[16]:= PartACollect[d_, k_, l_, ω_] :=
  PartA[d, k, l][All, 1] /. If[d == 2, OmegaRule2[ω, assu], OmegaRule3[ω, assu]]

In[17]:= FinalPoly[d_, k_, l_, sgn_] := Total[(PartA[d, k, l][All, 2] /. ω -> sgn) *
  ((PartACollect[d, k, l, ω] /. ω -> sgn) /. {Rulpl[d], Rulmin[d]}) * ListCoef[d, k, l][2] // Flatten]

In[18]:= UnrRepla = Fou[m_, n_] -> dHec[m] * Hec[n] - dHec[m - 1] * Hec[n - 1];

In[19]:= RamRepla = Fou[m_, n_] -> Hec[m + n] * Hec[m] - Hec[m + n + 1] * Hec[m - 1];

In[20]:= HecRecur[k_] := Hec[k] -> Hec[1] * Hec[k - 1] - A[2] * Hec[k - 2] + ω * Hec[k - 3];

In[21]:= AnyHecRec = Hec[k_] -> Hec[1] * Hec[k - 1] - A[2] * Hec[k - 2] + ω * Hec[k - 3];

In[22]:= GL3HecVar = Table[Hec[k - i], {i, 0, 4}];

In[23]:= (*The H-function for PGL(3)*)
Collect[SymmetricReduction[SymmetricReduction[R[3, 0, 0], {a[1], a[2], a[3]}, {A[1], A[2], ω}]]][1],
  {b[1], b[2], b[3]}, {B[1], B[2], ω}][1], x]

Out[23]=
1 + x^6 ω^4 - x^4 ω^2 A[1] × B[1] - x^2 A[2] × B[2] + x^3 (-2 ω^2 + ω A[1] × A[2] + ω B[1] × B[2])

In[24]:= (*Naive RS the first argument shifted by k*)

In[25]:= (*Unramified, PGL(2)*)

In[26]:= Collect[FinalPoly[2, k, 0, 1], x]

Out[26]=
x^2 Hec[-2 + k] - x B[1] × Hec[-1 + k] + Hec[k]

In[27]:= (*Ramified, PGL(2)*)

In[28]:= Collect[FinalPoly[2, k, 0, 0], x]

Out[28]=
Hec[k]

In[29]:= (*Unramified, PGL(3)*)

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In[30]:= FinUnrGL3 = Collect[FinalPoly[3, k, 0, 1], x]
Out[30]=

$$\begin{aligned}
& x^6 \text{Fou}[0, -3 + k] + \text{Fou}[0, k] - x^5 \text{B}[2] \times \text{Fou}[1, -3 + k] - \\
& x \text{B}[1] \times \text{Fou}[1, -1 + k] + x^4 \left( \text{B}[2]^2 \text{Fou}[0, -2 + k] + \text{B}[1] (-\text{Fou}[0, -2 + k] + \text{Fou}[2, -3 + k]) \right) + \\
& x^2 \left( \text{B}[1]^2 \text{Fou}[0, -1 + k] + \text{B}[2] (-\text{Fou}[0, -1 + k] + \text{Fou}[2, -2 + k]) \right) + \\
& x^3 (\text{Fou}[1, -2 + k] - \text{B}[1] \times \text{B}[2] \times \text{Fou}[1, -2 + k] - \text{Fou}[3, -3 + k])
\end{aligned}$$

In[31]:= FinUnrGL3InHec =
Collect[(FinUnrGL3 /. UnrRepla) /. {Hec[0] → 1, dHec[0] → 1, dHec[-1] → 0}, GL3HecVar]
Out[31]=

$$\begin{aligned}
& (x^5 \text{B}[2] - x^4 \text{B}[1] \times \text{dHec}[1] + x^3 \text{dHec}[2]) \text{Hec}[-4 + k] + \\
& (-x^3 + x^6 + x^3 \text{B}[1] \times \text{B}[2] - x^2 \text{B}[2] \times \text{dHec}[1] - x^5 \text{B}[2] \times \text{dHec}[1] + x^4 \text{B}[1] \times \text{dHec}[2] - x^3 \text{dHec}[3]) \text{Hec}[-3 + k] + \\
& (x \text{B}[1] - x^4 \text{B}[1] + x^4 \text{B}[2]^2 + x^3 \text{dHec}[1] - x^3 \text{B}[1] \times \text{B}[2] \times \text{dHec}[1] + x^2 \text{B}[2] \times \text{dHec}[2]) \text{Hec}[-2 + k] + \\
& (x^2 \text{B}[1]^2 - x^2 \text{B}[2] - x \text{B}[1] \times \text{dHec}[1]) \text{Hec}[-1 + k] + \text{Hec}[k]
\end{aligned}$$

In[32]:= (*Bounding*)
In[33]:= coeffsAndMonoms[FinUnrGL3InHec, GL3HecVar][[1]]
Out[33]=

$$\begin{aligned}
& \{1, x^2 \text{B}[1]^2 - x^2 \text{B}[2] - x \text{B}[1] \times \text{dHec}[1], \\
& x \text{B}[1] - x^4 \text{B}[1] + x^4 \text{B}[2]^2 + x^3 \text{dHec}[1] - x^3 \text{B}[1] \times \text{B}[2] \times \text{dHec}[1] + x^2 \text{B}[2] \times \text{dHec}[2], \\
& -x^3 + x^6 + x^3 \text{B}[1] \times \text{B}[2] - x^2 \text{B}[2] \times \text{dHec}[1] - x^5 \text{B}[2] \times \text{dHec}[1] + x^4 \text{B}[1] \times \text{dHec}[2] - x^3 \text{dHec}[3], \\
& x^5 \text{B}[2] - x^4 \text{B}[1] \times \text{dHec}[1] + x^3 \text{dHec}[2]\}
\end{aligned}$$

In[34]:= coeffsAndMonoms[FinUnrGL3InHec, GL3HecVar][[2]]
Out[34]=
{Hec[k], Hec[-1 + k], Hec[-2 + k], Hec[-3 + k], Hec[-4 + k]}
In[35]:= UnrGL3Expo =
Exponent[#, p] & /@ (List @@ Expand[#, x] & /@ coeffsAndMonoms[FinUnrGL3InHec, GL3HecVar][[1]] /.
{(Hec | dHec)[n_] → p^(n * Th), B[n_] → p^Th, x → p^(-1)})
Out[35]=

$$\begin{aligned}
& \{0, \{-2 + 2 \text{Th}, -2 + \text{Th}, -1 + 2 \text{Th}\}, \{-1 + \text{Th}, -4 + \text{Th}, -4 + 2 \text{Th}, -3 + \text{Th}, -3 + 3 \text{Th}, -2 + 3 \text{Th}\}, \\
& \{-3, -6, -3 + 2 \text{Th}, -2 + 2 \text{Th}, -5 + 2 \text{Th}, -4 + 3 \text{Th}, -3 + 3 \text{Th}\}, \{-5 + \text{Th}, -4 + 2 \text{Th}, -3 + 2 \text{Th}\}\}
\end{aligned}$$

In[36]:= UnrGL3RespScaleMax =
Assuming[0 < Th < 1/2, Max /@ (MapIndexed[#1 / #2[[1]] &, Rest[UnrGL3Expo]]) // FullSimplify]
Out[36]=

$$\left\{-1 + 2 \text{Th}, \frac{1}{2} (-1 + \text{Th}), \frac{2}{3} (-1 + \text{Th}), \frac{1}{4} (-3 + 2 \text{Th})\right\}$$

In[37]:= Assuming[5/14 < Th < 1/2, Max[UnrGL3RespScaleMax] // FullSimplify]
Out[37]=
-1 + 2 Th
In[38]:= (*Ramified, PGL(3)*)

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In[39]:= FinalPoly[3, k, 0, 0]
Out[39]=
Fou[0, k] - x B[1] × Fou[1, -1 + k] + x2 B[2] × Fou[2, -2 + k]

In[40]:= (FinalPoly[3, k, 0, 0] - Fou[0, k]) /. RamRepla
Out[40]=
-x B[1] (Hec[1] × Hec[k] - Hec[0] × Hec[1 + k]) + x2 B[2] (Hec[2] × Hec[k] - Hec[1] × Hec[1 + k])

In[41]:= FinRamGL3InHec = (FinalPoly[3, k, 0, 0] - Fou[0, k]) /. RamRepla /. {Hec[0] → 1, Hec[-1] → 0}
Out[41]=
-x B[1] (Hec[1] × Hec[k] - Hec[1 + k]) + x2 B[2] (Hec[2] × Hec[k] - Hec[1] × Hec[1 + k])

In[42]:= FinRamGL3InHecRec = Hec[k] + (FinRamGL3InHec /. HecRecur[k + 1] /. HecRecur[k] /. ω → 0)
Out[42]=
-x A[2] × B[1] × Hec[-1 + k] + x2 B[2] (Hec[2] (-A[2] × Hec[-2 + k] + Hec[1] × Hec[-1 + k]) -
Hec[1] (-A[2] × Hec[-1 + k] + Hec[1] (-A[2] × Hec[-2 + k] + Hec[1] × Hec[-1 + k]))) + Hec[k]

In[43]:= coeffsAndMonoms[FinRamGL3InHecRec, GL3HecVar][[1]]
Out[43]=
{1, -x A[2] × B[1] + x2 A[2] × B[2] × Hec[1] - x2 B[2] Hec[1]3 + x2 B[2] × Hec[1] × Hec[2],
x2 A[2] × B[2] Hec[1]2 - x2 A[2] × B[2] × Hec[2]}

In[44]:= coeffsAndMonoms[FinRamGL3InHecRec, GL3HecVar][[2]]
Out[44]=
{Hec[k], Hec[-1 + k], Hec[-2 + k]}

In[45]:= RamGL3Expo = Exponent[#, p] & /@
((List @@ Expand[#, p] & /@ coeffsAndMonoms[FinRamGL3InHecRec, GL3HecVar][[1]]) /.
{(Hec | A | B)[n_] → pn (n Th), x → p-1 (-1)})
Out[45]=
{0, {-1 + 3 Th, -2 + 5 Th, -2 + 5 Th, -2 + 5 Th}, {-2 + 6 Th, -2 + 6 Th}}

In[46]:= Assuming[0 < Th < 1 / 2, Max /@ (MapIndexed[#, #2[[1]] &, Rest[RamGL3Expo])] // FullSimplify
Out[46]=
{-1 + 3 Th, -1 + 3 Th}

In[47]:= (*Naive RS both arguments shifted by 1*)
In[48]:= (*GL2 Unr*)
In[49]:= Collect[FinalPoly[2, k, 1, 1], x]
Out[49]=
-x3 Hec[-1 + k] + B[1] × Hec[k] + x2 B[1] (Hec[-2 + k] + Hec[k]) + x (-B[1]2 Hec[-1 + k] - Hec[1 + k])

In[50]:= (*GL2 Ram*)
Collect[FinalPoly[2, k, 1, 0], x]
Out[50]=
B[1] × Hec[k]

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In[51]:= (*GL3 Unr*)

In[52]:= GL3UnrBoth1Gen = Collect[FinalPoly[3, k, 1, 1], x]

Out[52]=

$$\begin{aligned}
 & x^8 \text{Fou}[0, -1 + k] + B[1] \times \text{Fou}[0, k] + \\
 & x^7 B[2] (-\text{Fou}[0, -2 + k] - \text{Fou}[1, -1 + k]) + x (-B[2] \times \text{Fou}[0, 1 + k] - B[1]^2 \text{Fou}[1, -1 + k]) + \\
 & x^2 (B[1]^3 \text{Fou}[0, -1 + k] + \text{Fou}[0, 2 + k] + B[1] \times B[2] (-\text{Fou}[0, -1 + k] + \text{Fou}[1, k] + \text{Fou}[2, -2 + k])) + \\
 & x^6 (B[2]^2 (\text{Fou}[0, k] + \text{Fou}[1, -2 + k]) + B[1] (\text{Fou}[0, -3 + k] - \text{Fou}[0, k] + \text{Fou}[2, -1 + k])) + \\
 & x^3 (B[1]^2 B[2] (-\text{Fou}[0, k] - \text{Fou}[1, -2 + k]) + \\
 & \quad B[2]^2 (\text{Fou}[0, k] - \text{Fou}[2, -1 + k]) + B[1] (\text{Fou}[1, -2 + k] - \text{Fou}[1, 1 + k] - \text{Fou}[3, -3 + k])) + \\
 & x^4 (B[1] B[2]^2 (\text{Fou}[0, -2 + k] + \text{Fou}[1, -1 + k]) + B[1]^2 (-\text{Fou}[0, -2 + k] + \text{Fou}[0, 1 + k] + \text{Fou}[2, -3 + k]) + \\
 & \quad B[2] (-\text{Fou}[0, 1 + k] - \text{Fou}[1, -1 + k] + \text{Fou}[2, k] + \text{Fou}[3, -2 + k])) + x^5 (-B[2]^3 \text{Fou}[0, -1 + k] + \\
 & \quad \text{Fou}[1, k] + B[1] \times B[2] (\text{Fou}[0, -1 + k] - \text{Fou}[1, -3 + k] - \text{Fou}[1, k] - \text{Fou}[2, -2 + k] - \text{Fou}[3, -1 + k])
 \end{aligned}$$

In[53]:= GL3UnrBoth1 = Collect[GL3UnrBoth1Gen /. k → 1, x]

Out[53]=

$$\begin{aligned}
 & x^8 \text{Fou}[0, 0] + B[1] \times \text{Fou}[0, 1] + x^7 B[2] (-\text{Fou}[0, -1] - \text{Fou}[1, 0]) + x (-B[2] \times \text{Fou}[0, 2] - B[1]^2 \text{Fou}[1, 0]) + \\
 & x^2 (B[1]^3 \text{Fou}[0, 0] + \text{Fou}[0, 3] + B[1] \times B[2] (-\text{Fou}[0, 0] + \text{Fou}[1, 1] + \text{Fou}[2, -1])) + \\
 & x^6 (B[2]^2 (\text{Fou}[0, 1] + \text{Fou}[1, -1]) + B[1] (\text{Fou}[0, -2] - \text{Fou}[0, 1] + \text{Fou}[2, 0])) + x^3 \\
 & \quad (B[1]^2 B[2] (-\text{Fou}[0, 1] - \text{Fou}[1, -1]) + B[2]^2 (\text{Fou}[0, 1] - \text{Fou}[2, 0]) + B[1] (\text{Fou}[1, -1] - \text{Fou}[1, 2] - \text{Fou}[3, -2])) + \\
 & x^4 (B[1] B[2]^2 (\text{Fou}[0, -1] + \text{Fou}[1, 0]) + B[1]^2 (-\text{Fou}[0, -1] + \text{Fou}[0, 2] + \text{Fou}[2, -2]) + \\
 & \quad B[2] (-\text{Fou}[0, 2] - \text{Fou}[1, 0] + \text{Fou}[2, 1] + \text{Fou}[3, -1])) + \\
 & x^5 (-B[2]^3 \text{Fou}[0, 0] + \text{Fou}[1, 1] + B[1] \times B[2] (\text{Fou}[0, 0] - \text{Fou}[1, -2] - \text{Fou}[1, 1] - \text{Fou}[2, -1] - \text{Fou}[3, 0])
 \end{aligned}$$

In[54]:= Replsch = Fou[m_., n_.] → SchurPolynomial[{a[1], a[2], a[3]}, {m, n}];

In[55]:= ExceReplsch = (Fou[m_., n_.] /; TrueQ[m < 0 || n < 0]) → SchurPolynomial[{a[1], a[2], a[3]}, {m, n}];

In[56]:= GL3Unrsh1Repl = Collect[SymmetricReduction[
 GL3UnrBoth1 /. {ExceReplsch, Fou[0, 0] → 1}, {a[1], a[2], a[3]}, {Fou[0, 1], Fou[1, 0], 1}][[1], x]

Out[56]=

$$\begin{aligned}
 & x^8 + B[1] \times \text{Fou}[0, 1] - x^7 B[2] \times \text{Fou}[1, 0] + x (-B[2] \times \text{Fou}[0, 2] - B[1]^2 \text{Fou}[1, 0]) + \\
 & x^2 (B[1]^3 - B[1] \times B[2] + \text{Fou}[0, 3] + B[1] \times B[2] \times \text{Fou}[1, 1]) + \\
 & x^6 (-B[1] \times \text{Fou}[0, 1] + B[2]^2 \text{Fou}[0, 1] + B[1] \times \text{Fou}[2, 0]) + x^3 \\
 & \quad (-B[1] \times \text{Fou}[0, 1] - B[1]^2 B[2] \times \text{Fou}[0, 1] + B[2]^2 \text{Fou}[0, 1] + B[1] \text{Fou}[1, 0]^2 - B[1] \times \text{Fou}[1, 2] - B[2]^2 \text{Fou}[2, 0]) + \\
 & x^4 (B[1]^2 \text{Fou}[0, 2] - B[2] \times \text{Fou}[0, 2] - B[1]^2 \text{Fou}[1, 0] - B[2] \times \text{Fou}[1, 0] + B[1] B[2]^2 \text{Fou}[1, 0] + B[2] \times \text{Fou}[2, 1]) + \\
 & x^5 (2 B[1] \times B[2] - B[2]^3 + \text{Fou}[1, 1] - B[1] \times B[2] \times \text{Fou}[1, 1] - \text{Fou}[3, 0])
 \end{aligned}$$

In[57]:= **(*Bounding*)**

List @@ Expand[GL3Unrsh1Repl]

Out[57]=

$$\{x^8, x^2 B[1]^3, -x^2 B[1] \times B[2], 2 x^5 B[1] \times B[2], -x^5 B[2]^3, B[1] \times Fou[0, 1], -x^3 B[1] \times Fou[0, 1], \\ -x^6 B[1] \times Fou[0, 1], -x^3 B[1]^2 B[2] \times Fou[0, 1], x^3 B[2]^2 Fou[0, 1], x^6 B[2]^2 Fou[0, 1], \\ x^4 B[1]^2 Fou[0, 2], -x B[2] \times Fou[0, 2], -x^4 B[2] \times Fou[0, 2], x^2 Fou[0, 3], -x B[1]^2 Fou[1, 0], \\ -x^4 B[1]^2 Fou[1, 0], -x^4 B[2] \times Fou[1, 0], -x^7 B[2] \times Fou[1, 0], x^4 B[1] B[2]^2 Fou[1, 0], \\ x^3 B[1] Fou[1, 0]^2, x^5 Fou[1, 1], x^2 B[1] \times B[2] \times Fou[1, 1], -x^5 B[1] \times B[2] \times Fou[1, 1], \\ -x^3 B[1] \times Fou[1, 2], x^6 B[1] \times Fou[2, 0], -x^3 B[2]^2 Fou[2, 0], x^4 B[2] \times Fou[2, 1], -x^5 Fou[3, 0]\}$$

In[58]:= **BdListGL3UnrShf1[Th_] := Exponent[#, p] & /@**

((List @@ Expand[GL3Unrsh1Repl]) /. {x → p[^](-1), Fou[m_., n_] → p[^]((m + n) * Th), B[n_] → p[^]Th})

In[59]:= **Assuming[5 / 14 < Th < 1 / 2, Max[BdListGL3UnrShf1[Th]] // FullSimplify]**

Out[59]=

2 Th

In[60]:= **(*GL3 Ram*)**

Collect[FinalPoly[3, k, 1, 0], x]

Out[61]=

$$B[1] \times Fou[0, k] + x (-B[2] \times Fou[0, 1 + k] - B[1]^2 Fou[1, -1 + k]) + \\ x^2 B[1] \times B[2] (Fou[1, k] + Fou[2, -2 + k]) - x^3 B[2]^2 Fou[2, -1 + k]$$

In[62]:= **GL3Ramsh1Repl = ((Collect[FinalPoly[3, k, 1, 0], x]) /. k → 1) /. ExceReplsch**

Out[62]=

$$B[1] \times Fou[0, 1] + x (-B[2] \times Fou[0, 2] - B[1]^2 Fou[1, 0]) + x^2 B[1] \times B[2] \times Fou[1, 1] - x^3 B[2]^2 Fou[2, 0]$$

In[63]:= **BdListGL3RamShf1[Th_] := Exponent[#, p] & /@**

(List @@ Expand[GL3Ramsh1Repl] /. {x → p[^](-1), Fou[m_., n_] → p[^]((2 m + n) * Th), B[n_] → p[^](n * Th)})

In[64]:= **Assuming[5 / 14 < Th < 1 / 2, Max[BdListGL3RamShf1[Th]] // FullSimplify]**

Out[64]=

2 Th